# Information-Theoretic Semantics

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## Abstract and Keywords

Informational semantics takes the primary — at least the original — home of meaning to be the mind: meaning as the content of thought, desire, and intention. The meaning of beliefs, desires, and intentions is what it is we believe, desire, and intend. The sounds and marks of natural language derive their meaning from the communicative intentions of the agents who deploy them. As a result, the information of chief importance to informational semantics is that occurring in the transactions between animals and their environments. So for informational semantics the very existence of thought and, thus, the possibility of language depends on the capacity of (some) living systems to transform information (normally supplied by perception) into meaningful (contentful) inner states like thought, intention, and purpose.

Keywords: informational semantics, meaning, natural language, communicative intentions, perception, information

INFORMATION-THEORETIC semantics is an attempt to ground meaning—as meaning is understood in the study both of language and of mind—in an objective, mind- (and language-) independent notion of information. It is typically part of a larger philosophical effort to naturalize mentality—to understand thought and its expression in language as just another strand in the fabric of material affairs.

If symbols are the bearers of meaning, the things in the world that *have* meaning, information-theoretic semantics locates the primary source of this meaning in the relations these symbols bear to the world they are, or purport to be, about. These symbol–world relations can be described in explicitly information-theoretic terms (source, signal, noise, receiver, etc.), as occurs in Dretske (1981), or they can be described in more general causal terms (Stampe 1977, 1986; Matthen 1988; Fodor 1990; Israel and Perry 1990). However it may be expressed, though, the basic idea is that the primary determinant of a symbol's meaning is those situations in the world about which symbols carry, or are supposed to carry (more about this important qualification in a moment), information, those situations in the world of which the symbols are (in normal conditions) reliable signs.

Since, according to some (e.g. Dretske 1986; Israel and Perry 1990), the information an event carries is what its occurrence indicates, informational semantics is sometimes referred to as *indicator semantics*: what a symbol means is what, in certain central cases, it indicates, or is supposed to indicate, or normally indicates, about other parts of the world. Theories of this sort are to be contrasted with *conceptual-role, procedural*, or (p. 382) *consumer* semantics—theories that locate the meaning of a symbol not (or not *only*) in upstream causes, but also (and sometimes *primarily*) in the downstream effects these symbols have on other symbols and on the behaviour of the system in which they occur (Block 1986; Papineau 1987; Millikan 1989).

Informational semantics takes the primary—at least the original—home of meaning to be the mind: meaning as the content of thought, desire, and intention. The meaning of beliefs, desires, and intentions is what it is we believe, desire, and intend. The sounds and marks of natural language derive their meaning from the communicative intentions of the agents who deploy them. As a result, the information of chief importance to informational semantics is that occurring in the transactions between animals and their environments. So for informational semantics the very existence of thought and, thus, the possibility of language depends on the capacity of (some) living systems to transform information (normally supplied by perception) into meaningful (contentful) inner states like thought, intention, and purpose.

# 22.1 Information

The word ‘meaning’ is ambiguous. Two of its possible meanings (Grice 1989) are: (1) *non-natural* meaning—the sense in which the English word ‘fire’ stands for or means fire; and (2) *natural* meaning—the way in which smoke (not the word ‘smoke’, but smoke itself) means (indicates, is a sign of) fire. Non-natural meaning, the kind of meaning theories of meaning (like information-theoretic semantics) are supposed to be theories of, has no necessary connection with truth. The words ‘Jim has the measles’ mean that Jim has the measles whether or not Jim has the measles. Natural meaning, on the other hand, requires the existence of the condition meant: if Jim does not have the measles, the red spots on his face do not mean (indicate) that he has the measles. Perhaps all they mean is that he has been eating too much candy. Natural meaning, what an event indicates, what it is a sign of, is a relation between the sign and what it signifies that does not depend on anyone recognizing or identifying what is meant. It does not depend on conscious beings at all. Expanding metal (e.g. the mercury in a thermometer) indicates or means that the temperature is increasing whether or not anyone knows that this is what it means. A particular cloud formation means a cold front is approaching even if (because no one is aware of its meaning) it does not mean this *to* anyone. We found out that this is what it means. Unlike non-natural meaning, what things indicate, what they mean in a natural sense, is something we learn about events by patient investigation. We learn what tracks in the snow (or a cloud chamber) mean; we do not, as we do with words and other conventional symbols, assign them their meaning.

Information, as this is used in informational semantics, is akin to natural meaning. It is what the signal, some event occurring at a 'receiver' (an animal's brain, for instance), indicates about conditions existing at a causal 'source' (perhaps an object in the animal's environment). It might, for instance, mean (indicate, carry the (p. 383) information) that the object is moving, that it is located between two other objects, or that it is changing colour. The information carried by a signal, as so understood, is a fact—the fact expressed by the true proposition describing what the signal indicates about the source. So if *e*, an event at the receiver, indicates that *s*, an object at the source, is moving to the left, this fact about *s* is the information *e* carries about *s*. To speak about information is to speak about facts—the true propositions about a source that we (if properly atuned) can come to know about that source by receipt of a signal carrying that information. That, indeed, is why information is such an important commodity. It is why we, human beings, pay money (for tuition, for instance) to obtain it, why (in wartime) people are tortured in order to extract it from them, and why billions are spent on scientific instruments to obtain increasingly precise and exotic forms of it. The tendency in computer science to construe information as anything—whether true or false—capable of being stored on a hard disk is to confuse non-natural meaning (or perhaps just structure) with genuine information. It leaves it a mystery why information should be thought a useful commodity. Information booths are not merely places where one goes to hear people utter meaningful sentences. They are places you go expecting people to utter meaningful *true* sentences about the topics they dispense information about.

Thinking of information in terms of natural signs and what they indicate makes it clear that information depends on a system of stable relations existing between signal and source. It isn't enough, for instance, that *s*, an object at a source, always happens to be moving whenever a particular type of event, *e*, occurs at the receiver. That isn't enough to make *e* indicate that *s* is moving. No, for *e* to mean (in the natural sense) that *s* is moving, for *e* to carry this piece of information, *e* must depend on *s*'s movement in a particularly reliable way. Circumstances must be such that in these circumstances *only s*'s movement will result in *e*. If, in these conditions, something else can result in *e*, then the occurrence of *e* does not mean that *s* moved. The ringing of my telephone doesn't indicate, doesn't, therefore, carry the information, that *you* are calling me simply because, in point of fact, you are the only person who ever calls me. What is relevant is not whether anyone else ever *does* call me, but whether anyone else *might* call me—even if only someone dialing a wrong number or a bothersome telemarketer. If it might be someone else, then the ringing phone does not indicate that *you* are calling me. At most it means that it is probably you.

Another way of expressing this relation between source and receiver, the relation on which the flow of information depends (one I adopted in Dretske 1981 in order to make clear its connection with the mathematical theory of information and cognitive science in general), is to say that *e*, some event at a receiver, carries information about an object, *s*, at the source—the information, say, that *s* is F—only if conditions are such that *e* raises the probability of *s*'s being F to 1. It isn't enough to raise the probability to something less than 1—to 0.99 for instance. That isn't good enough. The reason it isn't good enough is that the relation in question (whether called information, indication, or natural meaning) is a transitive relation: if *a* indicates (means, carries the information) that *b*, and *b* indicates (means, carries the information) that *c*, then *a* must indicate (mean, carry the information) (p. 384) that *c*. If force on the restraining spring means that an electrical current is flowing in the circuit, and current flow in the circuit means there is a voltage difference, then force on the spring indicates a voltage difference. If Sally's expression indicates she is interested, and interest indicates (at least partial) understanding, then Sally's expression indicates (at least partial) understanding. This is why no signal can carry the information that water is freezing without carrying all the information carried by freezing water—that, for example, the temperature is at or below 32 °F. Setting the probability (required for the transmission of information) at anything less than 1 would not preserve this transitivity. It is not (in general) true that if the probability of *b*, given *a*, is (at least) 0.99, and the probability of *c*, given *b*, is also (a least) 0.99, then the probability of *c*, given *a*, is also (at least) 0.99. It may be less than 0.99. So if we are going to use probability at all to express the relations in question, those relations between source and receiver that enable events at the receiver to carry information about the source, we *must* express these as probabilities of 1. Nothing less than 1 maintains the transitivity of this indicator (natural meaning, informational) relation.

It may be thought that this sets the bar too high. In our messy unpredictable world probabilities seldom, if ever, reach a value of 1. If information requires a probability of 1, then we seldom, if ever, get information. This worry betrays a misunderstanding. The probabilities in question, whether they are probabilities of 1, 0.99, or 0.5, are assigned against a background of stable circumstances, circumstances that can, but are assumed not to, change for purposes of fixing what can happen. In normal circumstances the ring of my doorbell carries information: it indicates or means that someone is at my door. It does so despite the fact that in different conditions—when there is a short circuit in the wiring, when there is an infestation by poltergeists, when squirrels have been trained to push doorbell buttons—a doorbell can be made to ring without anyone (a person) pushing the doorbell button. But in normal circumstances, circumstances that include the actually existing wiring and the absence of poltergeists and trained squirrels, there is nothing else besides someone pressing my doorbell button that can make my doorbell ring. So in these circumstances the ring indicates that someone is at my door. It carries this information. It makes the probability of someone's being at my door 1. It makes the probability 1 whether or not we know it is 1. If it doesn't make the probability 1, if there is, unknown to us, a small probability that my doorbell is ringing when no one is at my door (there is a trained squirrel that ranges freely in the neighbourhood), then the ring does not indicate that someone is at my door. All it indicates is that someone is probably there. It might be the squirrel.

# 22.2 Meaning

Semantics is the study of meaning in so far as meaning is understood to be *non*-natural meaning: the kind of meaning in which ‘smoke’ means smoke (not fire); the (p. 385) kind of meaning in which a prophet can say, and thus mean, that the end is near without the end actually being near; the kind of meaning in which something in a person's head, for instance a perceptual experience or a belief, can mean (represent) one line to be longer than another even though the lines are of equal length. Information, as we have just described it (i.e. as indication or natural meaning), obviously cannot be equated with meaning in this (non-natural) sense. Nothing can carry the information that one line is longer than another, nothing can (naturally) mean or indicate that this is so, unless, in fact, the line is longer than the other. So an information-theoretic semantics, if it is to do the job, if it is to provide a plausible theory of non-natural meaning, must supplement or combine information as this is presently being understood with some other ingredient to capture the targeted concept.

There is some disagreement about what this additional idea or ingredient might be, but, whatever it is, it must somehow manage to convert natural signs of F, things that carry information about the F-ness of things, into things, symbols, which do not (or need not) carry such information. The favoured way of doing this is by appealing to the sort of ‘norms’ generated by things having a certain function. The basic idea is modelled on the way we give certain objects—measuring instruments, for instance—the power to mean, in a non-natural way, that something is so-and-so by giving these objects the job or function of ‘telling us’ (carrying the information) that something is so-and-so. A speedometer can say or represent that a car is going 60 mph when it isn't. What gives the instrument the power to mean, non-naturally, that the car is going 60 mph is the fact that it has a certain informational function (a function we—designers, makers, and users—give it): the function of telling us, providing us with information, about the speed of the car. When it is doing its job, it tells us how fast the car is going. If, because of mechanical difficulties, the instrument ceases to do its job, it nonetheless still ‘says’ (represents) that the car is going a certain speed, but it now misrepresents the speed. It (by pointing at the numeral ‘60’) says something false. It exhibits a form of non-natural meaning.

No one, however, gives our brains an information-carrying function in the way that we give measuring instruments their information-carrying functions. So if this is to be a model for how our brains acquire (in the form of experience and thought) the power to mean that something is so (whether or not it is so), the power to think that one line is longer than another when it isn't, the information-carrying functions must be (what we may call) natural functions—perhaps biological functions—functions that do not depend on or derive from us in any way. To smuggle in functions that, in whatever way, depend on our purposes in the way the functions of speedometers depend on them is to smuggle in at the very beginning of the analysis the very thing—the kind of meaning associated with human purposes and thought—that our theory of meaning is supposed to be an analysis of.

Setting aside for the moment questions about the existence of natural functions (I return to this in the next section), the idea would be to equate meaning or representational content with an element's information-carrying function. The functions convert natural signs into non-natural symbols. If event type E has tokens, (p. 386) $e_1, e_2, \ldots$ which, in normal circumstances, are natural signs that *s* is F, and if E acquires, through an appropriate history, the function of carrying this information, then tokens of that type thereafter mean, in a non-natural way, that *s* is F even when *s* isn't F. They mean (non-naturally) this because that is what they are supposed to mean (naturally). The norm invoked by speaking of this as something they are *supposed* to mean is given by their information-carrying function. They are supposed to mean this in the same way hearts are supposed to pump blood.

Just as a cashier, someone whose job is to operate the cash register, can be asked to sweep floors, a symbol for F, something whose job it is to indicate F, can be pressed into kinds of service unrelated to its information-carrying function. It might appear, for instance, in a desire for F, a fear of F, or just idle thought about F. Once we have a word for F in the language of thought, this word can now occur in mental questions (I wonder whether that is an F), commands (Give me an F), and hopes (I hope that is an F), as well as assertions (judgements that something is an F).

# 22.3 Problems and Promises

That is the good news. Now for some bad news. How bad the news is depends on how intractable the following problems are. I make suggestions here and there, but sometimes, with the customary excuse about lack of space, I provide nothing more than a promise about how an answer might go. The ultimate plausibility of information-theoretic semantics depends on there being detailed and convincing solutions to these problems.

## 22.3.1 Natural Functions

I spoke above of natural functions, the kind of function a thing has that is independent of anyone's recognition and/or attribution of that function. Stop signs and speedometers have functions, but they are not natural functions. As we all know, what stop signs and speedometers are supposed to do, or what we are supposed to do with them, derives from the collective purposes, desires, and beliefs of their designers, makers, and users. Information-based semantics needs something else. It needs natural functions, functions that (to avoid circularity) are independent of collective (or individual) purposes and beliefs. Where are these functions? What is it *in nature* that gives the visual systems of animals the function of providing information about optical surroundings? What makes the provision of information something the eyes, ears, and nose are for? Granted they *do* supply information; we couldn't survive without it. The question is whether there is any sense in which they are *supposed* to supply it.

The presumption is, of course, that the functions have their source in natural selection and learning. If the heart is supposed to pump blood and the liver is supposed (p. 387) to clean it, if these are the functions of these bodily organs created by a process of natural selection, why can't the eyes and ears and associated neural systems also have duties—in particular, information they are supposed to provide? Or if during operant conditioning, a pervasive form of learning, internal indicators of those stimulus conditions with which reinforced behaviour is to be coordinated come to exercise control over behaviour (the only way this kind of learning can be successful), why isn't it thereafter the job, the function, the purpose of these indicators (e.g. perceptual states) to provide information about the stimulus conditions on which the acquired behaviour depends?

It isn't important that the word ‘function’ be used to describe what these organs or systems develop to do. Some people seem to think that the word ‘function’ (in the sense of ‘purpose’, not in the sense of ‘causal role’) is only really appropriate when applied to systems to which human beings, given their special interests, have assigned a purpose.[1] According to this view, there is, independent of our interests, nothing the eyes, ears, and nose are for, nothing that they are supposed to be doing. We needn't, however, quarrel over the word ‘function’. What is important for the purposes of information-theoretic semantics is that there be a set of circumstances, or perhaps a kind of history, that, independent of human interests, grounds descriptions of animals and their parts as ill, sick, broken, damaged, injured, diseased, defective, flawed, infected, contaminated, or malfunctioning. If the truth of these descriptions is independent of *our* interests and purposes,[2] then there is a way natural systems are supposed to be, or supposed to behave, that is independent of how we conceive them. There would, therefore, be a perfectly natural sense in which, given appropriate development, the information-processing systems in animals would be able to make mistakes—able, that is, to mean that something was so when it wasn't. Misrepresentation, the sine qua non of non-natural meaning, would thus become possible in a perfectly natural way. It is hard to see why this much isn't available.

## 22.3.2 The Disjunction Problem

In order to mean, in the non-natural way we are trying to understand, that *s* is F, it must be possible for something to mean this when *s* isn't F. It doesn't have to be a dog for me to think and say it is. Symbols (linguistic or mental) that mean dog don't have to be caused by dogs. They can be caused by wolves or (Fodor's example[3]) cats-on-a-dark-night—things one might mistake for a dog. But anything that means or indicates, anything that carries the information, anything that is a sign, that *s* is a dog (p. 388) has to be caused by a dog.[4] This is the fundamental difference between natural and non-natural meaning.

This seems to create a problem. How can you convert something that is a sign of F, something that indicates that F, something that (carrying information about F-ness) can only be caused by an F, into something, a symbol for F, that can be caused by non-Fs?

Fodor (1990: 60) was certainly right that almost all attempts to solve this problem rely on distinguishing two contexts—one (call it the pure or normal context) in which the symbol for F is only caused by Fs (thus carrying information about the F-ness of a source) and one, call it the impure context, in which it can be caused by a variety of non-Fs (those, for instance that look like Fs), thus making misrepresentation (and, therefore, non-natural meaning) possible. The symbol type gets its information-carrying function, and thus its meaning, in the first context, when it is carrying information. This gives it a meaning that it takes to the impure context, a context in which, given its acquired function, it now means F (what it has the function of indicating) without anything necessarily being F.

The problem is to understand how ‘pure’ cases are possible, the conditions in which something can indicate dog while, at the same time, being capable of being caused (in the impure condition) by a wolf. If in impure conditions the symbol can be caused by a wolf (or anything else one might mistake for a dog), then why wouldn't it have been caused by a wolf in the pure case if, contrary to fact, a wolf had appeared there? If it would have been caused by a wolf had a wolf appeared in the pure condition, then how, in the normal or pure case, can the symbol indicate, how can it carry the information, that something is a dog? All it really seems to indicate, given the constraints on information described earlier, is a disjunctive (hence the ‘disjunction’ problem) fact, the fact that *s* is either a dog or a wolf. If that disjunctive fact is, indeed, the information it really carries, then the information the symbol acquires the function of carrying must be this disjunctive information. According to information-theoretic semantics, then, this disjunctive fact is what the element comes to (non-naturally) mean. If that is what it non-naturally means, however, then instances of the symbol caused by wolves in the impure condition are not misrepresentations. The cause—a wolf—actually is what the symbol's meaning says it is; namely, a wolf or a dog. So we have not achieved what we were after; that is, a symbol, something that can mean dog when it is caused by, and thus applied to, a wolf.

A lot of ink has been spilt on this problem, and nothing I can say in the space of a few paragraphs is going to settle matters, but I think it worth emphasizing that natural meaning, indication, and, therefore, information (of the relevant sort) are all context-dependent notions. Something can indicate F in one set of circumstances, (p. 389) not in another. Certain coloration and markings on the bird indicate that it is a blue jay, but these signs can (be made to) occur on an object that is not a blue jay. If circumstances are such that fake jays (non-blue jays that have the same colour and markings) are a genuine possibility, then coloration and markings of that sort do not indicate that the bird is a blue jay. They only indicate something disjunctive in character—that it is either a blue jay or a fake jay. Readers familiar with recent developments in epistemology will recognize these points as part of the ‘relevant-possibility’ theme in theories of knowledge. Recognizing a blue jay at the bird feeder does not require that one be able to distinguish real blue jays from fake jays—not if fake jays are not, as they usually aren't, relevant possibilities under normal viewing conditions. That is why knowing there is water (i.e. $H_2O$) in the creek does not require being able to distinguish $H_2O$ from XYZ (a molecularly distinct substance found on Twin Earth that looks like water). Not if XYZ is not, as it clearly is not if it is found *only* on Twin Earth, a relevant possibility. If, on the other hand, XYZ is actually found in some lakes and streams on earth, then you don't know the water you see is water even if it is water. It might, for all you can tell, be XYZ.

Quite independently of theories of meaning, then, we must recognize that something can carry information about dogs (blue jays, water) in one context, not in another. There are pure contexts, contexts in which dogs have to be taken as the only relevant cause of (the symbol) DOG, and impure contexts, contexts in which wolves, cats-on-a-dark-night, and perhaps even drugs in the bloodstream might be causing DOG. This being so, I see no problem in appealing to this distinction in a theory of meaning. A symbol acquires the function of carrying information in one context, a pure context, and it is used in contexts, impure contexts, in which it may or may not carry the information it has the function of carrying, contexts in which it can, therefore, get things wrong. My own view is that the pure contexts are those in which an information-carrying element is incorporated into a control system, a context defined by that time in which the information (relevant to an organism's adaptive behaviour) 'gets its hand on the steering wheel' (see Dretske 1981, 1988). But there are other possibilities.

## 22.3.3 The Grain Problem

Not only do information-carrying functions give physical systems (and, therefore, the central nervous system) the power to misrepresent the world—thus serving as one way natural meaning is converted into non-natural meaning—they also hold promise of solving a related problem that any naturalized semantics faces: the problem of *grain*. Non-natural meaning individuates propositions in a very fine-grained way. Thus, for instance, '*b* is 32 °F' means (non-naturally) something different from '*b* is 0 °C', even though these are merely two different ways of picking out the same temperature: the freezing point of water. In saying that *b* is 32 °F one does not say that *b* is 0 °C. One who believes that *b* is 32 °F does not necessarily believe that *b* is 0 °C. These are two distinct beliefs, distinct meanings, distinct mental contents. Yet (p. 390) if something is a sign (indicates, carries the information) that *b* is 32 °F it must be a sign (indicate, carry the information that) *b* is 0 °C. We may not know it carries this information, of course, but it nonetheless carries it. If the probability, given the signal, of *b* being 32 °F is 1, then the probability that *b* is 0 °C is also 1. Natural meaning does not carve things up as finely as does non-natural meaning. This appears to be a problem for theories like information-theoretic semantics that propose to analyse non-natural meaning in terms of natural meaning. Non-natural meaning and natural meaning have different grains.

And so they do, but information-theoretic semantics is not making the mistake of identifying non-natural meaning with natural meaning. Non-natural meaning is being identified with an object's information-carrying function, and it is reasonably clear that information-carrying functions can carve up propositional space in a more finely grained way than does information. Of the many things a heart does, only one of these, pumping blood, is its biological function. So too, perhaps, although an internal state is indifferent to how the facts it carries information about are described (that *b* is 32 °F or 0 °C), it may have the function of carrying this information in terms of a value on a Fahrenheit rather than a centigrade scale. Enc (1982) gives the example of a mechanical device that, given the way it does its job (by counting lines, not angles), has the function of detecting trilaterals, not triangles, even though, geometrically speaking, these are the same. If it is possible for functions to pull apart, in this way, facts that are, informationally speaking, equivalent, non-natural meanings and informational functions could exhibit the same propositional grain.[5] There would therefore be no principled objection to equating non-natural meaning, as this appears in the form of thought and intention, with information-carrying functions even though information and meaning (of the sort being analysed) have very different grains.

# 22.4 Epiphenomenalism

If there really are semantic engines, systems whose behaviour (some of it anyway) is driven (explained) by the semantic properties of their internal states, a theory of meaning should reveal just how meaning gets its hands on the steering wheel. How does meaning achieve this causal relevance? If the neurobiological properties of internal events are not enough to explain why we do some of the things we do—the purposeful actions—how do the semantic properties, the content or meaning of internal events (i.e. what we believe, intend, and desire), figure in causal explanations of behaviour?

(p. 391)

The problem of finding a genuine explanatory role for meaning in the behaviour of intentional (i.e. semantic) systems is especially difficult for semantic theories that locate the source of meaning in the extrinsic or relational properties of the internal events that presumably have meaning. If a theory says that the meaning or content of internal events derives from the way these symbols are related—functionally, informationally, causally, or whatever—to (usually) external affairs, then it should be possible for physically indistinguishable events, symbols that have all their intrinsic properties in common, to have quite different meanings. This should be possible in the same way it is possible for identical twins, say, to have different histories. According to orthodox thinking about causality, however, the causally relevant properties of an object are local or intrinsic. Physically indistinguishable objects (a real $100 bill and a perfect forgery, say), placed in the same circumstances, will have exactly the same effects on the rest of the world. They will have different histories, but these historical (extrinsic, relational) differences will be masked or screened off by their current physical constitution. And so it is with meanings in so far as meaning supervenes on history. What will be causally relevant to the effects of symbols in a system will not be their history—and, therefore, their meaning. It will be their physical (intrinsic) properties.

Information-theoretic semantics has this problem in spades, since it locates the meaning of a symbol in the history of that symbol, in what kind of information it was, at some point in the past, developed to carry, and history is obviously not a local, not an intrinsic, and, therefore, not (according to orthodox theory) a causally relevant property. Every other semantic theory (and this seems to be most theories) that locates the meaning of a symbol in its extrinsic or relational properties has the same problem. This problem is most often dramatized by Davidson's swampman example. Lightning strikes a swamp creating a duplicate of Donald Davidson, a being that is physically indistinguishable from Davidson but one having a completely different history. Does one's theory of meaning assign the same meanings, the same content, to the internal states of this duplicate being as it does to Donald Davidson? If so, then history is irrelevant to the assigned meanings. If not, then the duplicate has no thoughts. It is a zombie. The second choice seems to be inconsistent with scientific materialism. The first choice is inconsistent with information-theoretic semantics. Conclusion? Information-theoretic semantics is inconsistent with scientific materialism. This wouldn't be so bad for some theories, but for information-theoretic semantics it is a disaster, since it was proposed as a materialistically acceptable account of meaning.

Some philosophers (e.g. Paul Churchland 1981; Patricia Churchland 1986) take this problem to be insurmountable and construe semantic explanations of behaviour, folk-psychological explanations couched in terms of the content or meaning of internal events, as, at best, a *façon de parler*, a convenient verbal heuristic that will eventually (when we know enough) be replaced by scientific (neurobiological) explanations of behaviour. Others (e.g. Dennett 1987) acknowledge the instrumental or heuristic nature of semantic explanations, but take the heuristic—the intentional stance—to be unavoidable. Still others argue for a genuine relevance (p. 392) for meaning in a variety of different ways: Fodor (1987) identifies an alleged kind of meaning, narrow meaning, that is intrinsic to the events that have it; Dretske (1988) argues that the behaviours to be explained by meaning are not the bodily movements that are best explained by neurobiology, but, rather, causal processes that result in bodily movements (which are best explained by the extrinsic properties of the internal causes); Kim (1996) identifies a kind of causality, supervenient causality, that extrinsic properties (and, therefore, meanings) can participate in; Burge (1989) defends the causal relevance of semantic properties by arguing that genuine causal explanations are those that are actually given and accepted in our daily explanatory practice (a practice loaded with semantic explanations).

Perhaps, though, the best defence against the charge of epiphenomenalism is to say that although the problem is real enough, it is not just a problem for information-theoretic semantics. It is a problem for any theory of meaning—and this seems to be *all* theories of meaning—in which meaning resides in a symbol's extrinsic or relational properties. In philosophy everybody's problem isn't anybody's problem.

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## Notes:

(1) For an excellent collection of essays debating this issue see Ariew et al. (2002).

(2) Is an animal sick or diseased, or is its leg broken, only if we, human beings, deem it so? That doesn't sound right. If these conditions really are independent of our (or, indeed, anyone's) beliefs and interests, then there is a way things are supposed to be that is, in the appropriate sense, natural or objective in the way required by information-theoretic semantics.

(3) It was Fodor who first raised the disjunction problem (1984). See also Fodor (1990).

(4) This isn't strictly true, since a signal can carry the information that *s* is F without being caused by *s*'s being F. There are causal arrangements (e.g. common causes) in which *e* carries the information that *s* is F without being caused by *s*'s being F. The claim in the text, though, is close enough to being true for the purposes at hand. Nothing can indicate or mean (naturally) that *s* is a dog, nothing can carry this information about *s*, unless, in fact, *s* is a dog.

(5) See Dretske (1981: 215–22) for a similar discussion of how to distinguish, on information-theoretic grounds and compositional considerations, the concept (and, therefore, belief that something is) WATER from the concept (belief that something is) $H_2O$ despite the (metaphysically) necessary equivalence of water with $H_2O$.

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